BCO570A **2 Credits**	***Digital, Data & AI Literacy***	**0:0:2**

Course Objective

1. To build foundational digital, data and AI literacy through a fully hands-on, lab-driven curriculum (10 minutes theory + 90 minutes practical per session).
2. To enable students to use and compare leading AI tools such as ChatGPT, Claude, Gemini and Perplexity for research, writing, prompting and productivity tasks.
3. To help students build a complete digital presence, including a personal website, resume, LinkedIn profile and AI chatbot, using no-code AI tools.
4. To develop practical skills in AI-driven visual, audio, video content creation, data analytics, automation and business intelligence.
5. To apply AI knowledge and tools in designing, building and presenting a real capstone product, from ideation through deployment and demo.

Course Syllabus — 24 Sessions

Module	Practical Exercise #	Exercise Details
Module-1: ***AI Foundations, Prompting & Productivity Tools***	1	Understanding AI — the only dedicated theory session: AI vs ML vs Deep Learning vs GenAI, how AI models learn, types of AI tools, hallucinations, bias and ethics, and India's AI ecosystem.
	2	Live comparison of ChatGPT, Claude, Gemini & Perplexity on the same tasks; build a personal AI Model Cheatsheet; learn the Role + Context + Task + Format + Constraints prompting framework and build a Prompt Portfolio.
	3	Use NotebookLM to generate structured notes, an audio podcast and Q&A flashcards from a textbook chapter; use Perplexity AI for a cited research brief; build a 10-slide presentation using Gamma.app.
	4	Use Claude and ChatGPT for professional writing (emails, reports, LinkedIn posts, cover letters); build a personal Notion AI workspace; use Gemini in Gmail and Google Docs AI for productivity.
	5	Build a personal website using Lovable.dev and Antigravity with no-code AI tools; customise About, Projects and Contact sections and obtain a live URL.
Module-2: ***Digital Presence, AI Product & Visual Content***	6	Deploy the website via GitHub and Vercel; register a domain and connect it via DNS; set up Google Search Console for indexing and discoverability.
	7	Use Claude and Kickresume to build an ATS-optimised resume; optimise a LinkedIn profile using ChatGPT; build a reusable Claude prompt that generates tailored cover letters.
	8	Build and train a custom AI chatbot on Botpress for a school-specific use case, train it on an uploaded PDF, and embed it on the personal website built in Session 5.
	9	Ideate, validate and build a complete AI product from idea to a live deployed URL using Lovable.dev/Bolt.new (Milestone 2: Website + Resume + LinkedIn + Chatbot + AI Product URL).
	10	Generate and progressively refine AI images using Adobe Firefly and Canva AI (Magic Design, Text-to-Image, Magic Write); edit photos using Remove.bg and generative fill.
Module-3:	11	Clone a voice and generate a narrated audio piece using ElevenLabs; edit a short video reel with AI captions and transitions using CapCut AI; generate an AI avatar presenter video using D-ID.

Module	Practical Exercise #	Exercise Details
AI Media, School Labs & Data Analytics	12	Build a full AI-generated pitch deck using Gamma.app; write formulas and auto-generate charts in Google Sheets with Gemini; generate infographics and flowcharts using Napkin.ai.
	13	School-specific AI tools lab: each school (Sciences, Journalism & Mass Comm, Humanities, Hospitality & HM, Design, BBA) applies domain-specific AI tools to a real deliverable.
	14	Analyse a provided dataset using Julius AI and ChatGPT Code Interpreter to generate insights, charts and correlations; build a live dashboard in Google Looker Studio.
	15	Analyse social media and website metrics using Metricool and Google Analytics 4; build an AI-suggested content calendar; run a brand sentiment analysis using ChatGPT and Perplexity.
Module-4: ***AI Automation, Business Intelligence, Careers & Safety***	16	Build no-code automations using Zapier AI and Make.com; auto-generate structured meeting notes with Notion AI (Milestone 3: Looker Studio dashboard + 3 live Zapier automations).
	17	Build a personal finance report and budget planner using Julius AI and ChatGPT; run a business case analysis (SWOT, revenue projection) and build a pitch deck with Gamma.app; use Claude for structured decision-making.
	18	Research AI's impact on a chosen career field using Perplexity; build a personal 3-year AI-augmented career plan with Claude; run a resume-to-job-description skill-gap analysis; explore AI freelancing gigs.
	19	Detect deepfakes in mixed real/AI-generated media using Hive Moderation and FotoForensics; audit personal AI/social privacy settings; fact-check viral claims using Perplexity and Google Fact Check Explorer.
	20	Advanced school-specific AI lab building on domain tools from Session 13; Capstone briefing — project brief, team formation, problem-statement submission and tool selection.
Module-5: ***Capstone — Build Something Real with AI***	21	Scope the capstone problem using Perplexity research; write a one-page project brief with Claude (problem, target user, AI solution, tools, success metric); present for faculty approval and set up the project board.
	22	Full build session: construct the core AI product/prototype using Lovable.dev/Bolt.new, Botpress, Gamma.app or Looker Studio as applicable, with faculty mentoring and a cross-team demo.
	23	Refine the product based on feedback; build a final 10-slide Gamma.app deck; record a demo video; conduct structured peer review and incorporate at least two peer suggestions.
	24	Final Showcase & Viva — each team presents and demonstrates its AI product to a faculty panel followed by individual viva questions; top projects recognised and featured on the JECRC AI Showcase page.

Course Outcome

CO-1	Student understands the foundations of AI (AI vs ML vs Deep Learning vs GenAI), compares leading AI tools, applies structured prompting frameworks, and uses AI for research, notes and professional writing.
CO-2	Student builds and deploys a complete personal digital presence — a live website, ATS-optimised resume, LinkedIn profile, a trained AI chatbot and a deployed AI product — using no-code AI tools.
CO-3	Student creates AI-generated audio, video and presentation content, applies domain-specific AI tools relevant to their own school, and performs data analytics using AI.
CO-4	Student automates workflows, builds AI-driven business and finance reports, plans an AI-augmented career path, and practices safe, responsible AI use.
CO-5	Student designs, builds and presents a complete capstone AI product — from problem scoping and prototyping to a public demo and faculty viva.

CO-PO-PSO Mapping Table (H: High, M: Medium, L: Low)

CO/PO-PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO-1	H	M			M	H	M	L	M	M		M	M		M
CO-2		H		M	M		M		M	H			H	M	
CO-3	M	M	H			M					L		M	M	L
CO-4	H	H	H	H	H	H	H		L	H	H	H	H	H	
CO-5	H	H	H	H	H	M	H	L			L	H	H	H	H